

**Midterm Exam**  
Logic and Sets for AI

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*This exam has **3 pages** and **7 exercises** with a total of **80 points**. The result will be computed as  $1 + \left(\frac{\text{total}}{80} \cdot 9\right)$*

*Note that in this course, 0 is considered a natural number. If you are, in any exercise, unsure about the definition of some term/operator/... state your assumed definition at the beginning of the exercise.*

*Please put **your name on every sheet** you hand in, and write as clearly as possible. Submissions (even parts) without a name might **not be considered**. If we cannot read a statement it will be considered **incorrect**.*

Good luck!

**1. Semantic entailment of Sheffer Stroke (10 points)**

Determine if the following semantic entailment holds:

$$(p \rightarrow (q \oplus r)) \wedge ((q \mid \neg r) \vee p) \models \neg q \leftrightarrow r$$

**2. CNF (4 + 4 + 2 points)**

- (a) Construct a formula in CNF that is semantically equivalent to a formula

$$(p \vee r) \vee (\neg(p \wedge q) \rightarrow r)$$

using **the truth table method**.

- (b) Determine a semantically equivalent CNF of the formula

$$\neg((p \vee q) \rightarrow r) \wedge (\neg p \rightarrow r)$$

by **applying the algorithm CNF**.

- (c) Explain and apply the criterion for determining whether a CNF is a tautology to the CNF you constructed in (b).

**3. DPLL (9 points)**

Apply the DPLL procedure to the CNF

$$(p \vee \neg r) \wedge (q \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (q \vee r)$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Unless there is unit clause or a pure literal, let the procedure branch **first on p, then on q, and then on r**; and let it always try the truth value true ( $\top$ ) before false ( $\perp$ ).

If the formula is satisfiable, please provide a model.

**4. Island Puzzle** (6 + 5 points)

- (a) On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders Xander, Yara and Zain.

Xander says: “Yara is a liar”

Yara says: “I am the same as Zain”

Zain says: “Xander is a truth speaker”

Determine by means of logical reasoning or by use of a truth table whether islanders X, Y and Z are truth speakers or liars.

- (b) On this same island, you are now facing two different islanders, Abraham and Bethel knowing that one of them is a truth speaker and the other is a liar but not knowing who is who. A road bifurcates in front of you, one path leading to a treasure and the other leading to peril. Both islanders know each other’s nature and the characteristics of the paths. With one clever question yes or no question that you might ask any one of them, you can be sure to find the safe path. What is the question and what is the direction which you would then take?

**5. Sets** (4 + 9 points)

Consider the following set-theoretic equality:

$$(A \cap B) \cap (A \Delta B) = \emptyset \quad (1)$$

- (a) Prove this identity using a sequence of Venn diagrams. You can draw a series of diagrams showing how one side of the equality is constructed from A and B via elementary set operations and use this to prove that the equation holds for any two sets A and B. Please label the diagrams so that it is clear which elementary operation or formula they correspond to.
- (b) Prove the identity using the algebra of sets. Please make sure to name all the rules which you apply. You may name more than one rule per line if it is clear and correct how you apply them.

**6. Relations** (4 + 6 + 6 points)

Given the set  $V = \{n, o, p, q, r, s\}$  with the relations

$$A = \{\langle n, p \rangle, \langle n, s \rangle, \langle o, o \rangle, \langle p, n \rangle, \langle q, r \rangle, \langle q, s \rangle\}$$

$$B = \{\langle n, n \rangle, \langle n, q \rangle, \langle q, r \rangle\}$$

- (a) Given the relation A, draw the directed graph representation.
- (b) Is the relation A reflexive? Is A transitive? Is B anti-symmetric? Why or why not?
- (c) Draw a Venn diagram depicting the composition  $A^{-1} \circ B$  and write down the matrix representation of it.

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**7. Ordering relations** (7 + 4 points)

Let  $P$  be a subset of  $\{\text{Morning}, \text{Afternoon}, \text{Evening}, \text{Night}\}^2$  and be defined as (with elements abbreviated by their first letters):

$$P := \{\langle M, M \rangle, \langle M, A \rangle, \langle A, M \rangle, \langle A, A \rangle, \langle E, E \rangle, \langle E, N \rangle, \langle N, E \rangle\}$$

Consider the Cartesian ordering relation on  $P$  which is induced by the order in which the times of day follow each other, i.e.,  $M \leq A \leq E \leq N$ .

- (a) Construct and draw the Hasse diagram of this Cartesian order on  $P$ . You can use the algorithm which you learnt about in class, but you are not required to do so. For the sake of simplicity, you can abbreviate pairs such as  $\langle M, A \rangle$  by writing  $MA$ .
- (b) Determine and list all the minimal and maximal elements in  $P$  according to this Cartesian ordering. Is there a minimum and/or maximum? If yes, what are these?

**End of the exam.**